PROBABILITY THEORY

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for Data-Driven Decision Making in Africa

Learning outcomes

- Understand the basic concepts of probability
- Understand the difference between unconditional and conditional probability
- Understand the different types of events and rules of probability
- Perform basic probability calculations

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Basic Ideas of Probability

- Probability is a concept used every day and in many different settings.

- *Objective (frequentist) probability*
  - The number of times an outcome occurs divided by the total number of trials (repetitions). An outcome that always occurs has a probability of 1; an outcome that never occurs has a probability of 0

- *Subjective probability*
  - One's degree of belief in the occurrence of an event

	Interpretation	Example
Objective (Frequentist)	Based on long-run frequency of events	100 of 1,000 people exposed to SARS-CoV-2 become infected (10%)
Subjective	Based on personal belief or expert opinion	An epidemiologist believes there is a 70% chance of a new outbreak next month
Bayesian	Updates beliefs using new evidence	A patient's probability of having COVID-19 increases from 20% to 95% after a positive test result

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Basic Ideas of Probability

- The probability of a particular outcome A is written as P(A)

- If a process is repeated n times, and an event A occurs m times

$$P(A) = \frac{m}{n}$$

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Probability calculations

- The table below gives the number of definite TB meningitis cases by CD4 count

CD4 Count	Def TB	Non TB	Total
< 50	9	11	20
50-200	22	20	42
≥ 200	8	23	31
Total	39	54	93

What is the probability that a randomly selected person from the study has definite TB Meningitis ?

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Probability Calculations

$$
\begin{aligned}
P(DefTB) &= \frac{\text{number of TB meningitis cases}}{\text{total number of subjects}} \\
&= \frac{39}{93} \\
&= 0.42
\end{aligned}
$$

What is the probability that a randomly selected person from the study has CD4 count less than 50?

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Probability Calculations

CD4 Count	Def TB	Non TB	Total
< 50	9	11	20
50-200	22	20	42
≥ 200	8	23	31
Total	39	54	93

$$P(CD4\ count < 50)$$

$$= \frac{number\ with\ CD4 < 50}{total\ number\ of\ subjects} = \frac{20}{93}$$

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Conditional Probability

The probability we just computed was an “unconditional” probability , since the size of the total sample served as the denominator. When probabilities are calculated with a subset of the total sample in the denominator we call them conditional probabilities.

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Conditional Probability

Example: If we randomly select a person from the 93 subjects and find that he has CD4 count < 50, what is the probability that this person will have TB meningitis?

CD4 Count	Def TB	Non TB	Total
< 50	9	11	20
50-200	22	20	42
≥ 200	8	23	31
Total	39	54	93

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Conditional Probability

- Total number of subjects no longer of interest, since with the selection of CD4 < 50 all other categories are eliminated

$$
\begin{aligned}
P(DefTB \mid CD4 < 50) &= \frac{\text{number of TB meningitis cases with } CD4 < 50}{\text{total number of subjects with } CD4 < 50} \\
&= \frac{9}{20} \\
&= 0.45
\end{aligned}
$$

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Conditional Probability

Formally,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

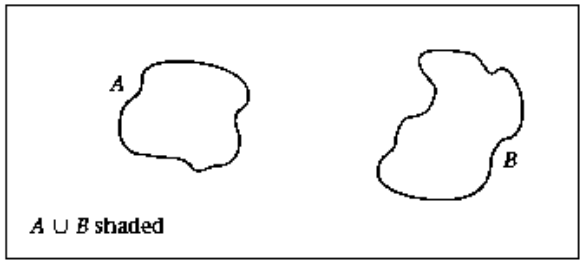
where $\cap$ denotes an intersection or and.
If we randomly pick a person from the 93 subjects and find that he has TB meningitis, what is the probability that this person will have $CD4 \geq 200$?

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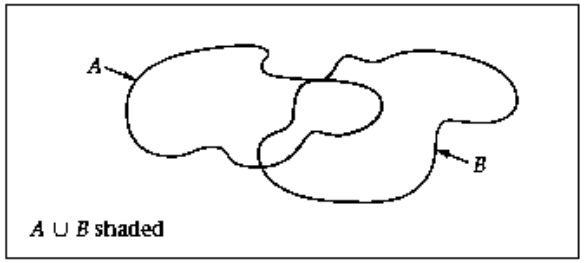
Mutually Exclusive Events: Addition Rule

- The probability of two mutually exclusive events occurring is the probability that either one event or the other event occurring.

Diagrammatic representation of $A \cup B$: (a) A, B mutually exclusive; (b) A, B not mutually exclusive



(a)



(b)

Mutually Exclusive Events: Addition Rule

The probability of two mutually exclusive events occurring is the probability that either one event or the other event occurring.

Table: Distribution of blood type by gender

Blood Type	Males	Females	Total
O	0.21	.021	0.42
A	0.215	0.215	0.43
B	0.055	0.055	0.11
AB	0.02	0.02	0.04
Total	0.5	0.5	1

For example, the probability that a randomly selected person has either blood type O or blood type A is

P(O or A) = P(O) + P(A) = 0.42 + 0.43 = 0.85

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Independent Events: Multiplication Rule

- Two events A and B are independent, then the probability of A is the same regardless of whether B occurred or not.

$$P(A|B) = P(A)$$

$$P(A \cap B = P(A)P(B)$$

Independent Events

Back to the blood type example:

Table: Distribution of blood type by gender

Blood Type	Males	Females	Total
O	0.21	.021	0.42
A	0.215	0.215	0.43
B	0.055	0.055	0.11
AB	0.02	0.02	0.04
Total	0.5	0.5	1

The probability of being male and having blood type O is

$$\begin{aligned} P(\text{male and blood type O}) &= P(\text{male}) \times P(\text{blood type O}) \\ &= 0.50 \times 0.42 \\ &= 0.21 \end{aligned}$$

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Some Applications of Probability

- Relative Risk
- Screening Tests

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Relative Risk

- The relative risk of a disease is the ratio of the probability of disease in people with the risk factor (exposed persons) to the probability of disease in people without the risk factor (non-exposed persons). The relative risk (*RR*) of *B* given *A* is

$$\mathrm{RR} = \mathrm{P}(\mathrm{B}|\mathrm{A})/\mathrm{P}(\mathrm{B}|\overline{A}).$$

Relative Risk

Example: Suppose that among 100,000 women with negative mammograms 20 will be diagnosed with breast cancer within 2 years, or $\mathrm{P}(B|\bar{A}) = 20\ /10\text{^}5\ = 0.0002$, whereas

1 woman in 10 with positive mammograms will be diagnosed with breast cancer within 2 years, or$P(B|A) = 0.1$.

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Relative Risk

$$RR = \frac{P(B|A)}{P(B|\bar{A})} = \frac{0.10}{0.0002} = 500$$

- In other words, women with positive mammograms are 500 times more likely to develop breast cancer over the next 2 years than are women with negative mammograms

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Screening Tests

- Screening is conducting a test on individuals who have not yet presented any symptoms in order to classify them with the probability of having a particular disease

- The following can happen

Marker	Present	Absent	Total
Positive	a (true positive)	b (false positive)	a+b
Negative	c (false negative	d (true negative	c+d
Total	a+c	b+d	Total

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Screening Tests: Sensitivity and Specificity

- *Sensitivity*
  - the probability that the test is positive given that the person has a disease or

$$P(test\ positive|disease)$$

- *Specificity*
  - the probability that the test is negative given the person does not have the disease or

$$P(test\ negative|no\ disease)$$

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Screening Tests: Sensitivity and Specificity

Example:

The chance of containment of cervical cancer is high if detected early. A Pap smear is a widely accepted screening procedure that can detect asymptomatic cancer. An on site proficiency test conducted in 1972, 1973 and 1978 assessed the competency of clinicians who scan Pap smear slides for abnormalities.

Overall 16.25% of tests performed on women with cancer resulted in false negative outcomes.

$$P(test\ negative|cancer) = 0.1625$$

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Screening Tests: Sensitivity and Specificity

$$P(test\ negative|cancer) = 0.1625$$

The other 83.75% tested did in fact test positive.

$$P(test\ positive|cancer) = 1 - 0.1625 = 0.8375$$

This is the sensitivity of the test.

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Screening Tests: Sensitivity and Specificity

Not all the women tested had cervical cancer. In fact 18.64% of the tests were false positive outcomes.

$$P(test\ positive|no\ cancer) = 0.1864$$

Therefore

$$P(test\ negative|no\ cancer) = 1 - 0.1864$$
$$= 0.8136$$

This is the specificity of the test.

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